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FOR THE MASTER'S DEGREE IN BIOINFORMATICS APPLIED TO PERSONALIZED MEDICINE AND HEALTH

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MADRID, SPAIN

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Summary

Hepatoblastoma (HB) is an uncommon liver cancer frequently diagnosed in children less than 5 years old, with a largely unknown origin. The most common alterations found in HB are activating mutations of the β-catenin encoding gene, with an incidence of 70%. In this study, we characterized the current landscape of high-throughput transcriptomics data in HB through a meta-analysis of 5 publicly available RNA-Seq studies, including a total of 181 samples, 101 patients, from which we identified commonly dysregulated genes in HB, 199 up- and 124 down-regulated. In order to study the impact of gene dysregulation on cellular functions, we constructed a multilayer network representing various levels of molecular information, defining metabolic reactions, protein-protein interactions in the liver, and pathways, that we used to uncover the key genes involved in the spreading of dysregulation, and therefore, identify novel candidate cancer genes for HB. This was accomplished, as supported by previous work, by detecting gene communities, allowing us to uncover 4 dysregulated genes with a strong impact on the system, namely LRRC52, DLX6, DLX5, and MSX2. We found that these genes could be linked to the pathway of β-catenin, and therefore to HB, and also connected to adult liver cancer. However, further experimental characterizations should be performed to confirm the involvement of the dysregulation of these genes in the disease. Moreover, to enable other scientists to use this workflow, we deployed it as a code-free tool to the Virtual Research Environment (VRE) of the European H2020 project iPC (ref. 826121).

Resumen

El hepatoblastoma (HB) es un cáncer de hígado poco común que se diagnostica normalmente en niños menores de 5 años, con un origen en gran parte desconocido. Las alteraciones más comunes encontradas en HB son mutaciones activadoras del gen que codifica la β-catenina, con una incidencia del 70%. En este estudio, caracterizamos el panorama actual de datos de transcriptómica de alto rendimiento en HB a través de un metanálisis de 5 estudios RNA-Seq disponibles públicamente, incluyendo un total de 181 muestras, 101 pacientes, de los cuales identificamos genes comúnmente desregulados en HB, 199 regulados al alza y 124 regulados a la baja. Para estudiar el impacto de la desregulación genética en las funciones celulares, construimos una red multicapa que representa varios niveles de información molecular, definiendo reacciones metabólicas, interacciones proteína-proteína en el hígado y vías, que usamos para descubrir los genes clave involucrados en la propagación de la desregulación y, por lo tanto, identificar nuevos genes de cáncer candidatos para HB. Esto se logró, como lo respaldan trabajos anteriores, mediante la detección de comunidades de genes, que permiten descubrir 4 genes desregulados con un fuerte impacto en el sistema, a saber, LRRC52, DLX6, DLX5 y MSX2. Descubrimos que estos genes podrían estar relacionados con la vía de la β-catenina y, por lo tanto, con la HB, y también con el cáncer de hígado en adultos. Sin embargo, se deben realizar más caracterizaciones experimentales para confirmar la participación de la desregulación de estos genes en la enfermedad. Además, para permitir que otros científicos utilicen este flujo de trabajo, lo implementamos como una herramienta sin código en el Virtual Research Environment (VRE) del proyecto europeo H2020 iPC (ref. 826121).

Contents

Chapter 1

Introduction

1.1 Biological systems and multilayer networks

Background

Over the years, the study of biology has focused on identifying individual genes, proteins, and cells, as well as studying their functions [1]. This perspective provides limited information about the functioning of living systems and a more holistic approach is being pursued in many different areas of computational biology, in particular the area of network medicine.

Since the turn of the century, we have progressed in several disciplines. Including molecular biology, bioinformatics, engineering, and physics. Whilst generating a wide spectrum of large-scale biological data [2]. There is an increasing demand for a suitable framework to comprehend and integrate these complex and dynamic datasets, such as integrating the interactions between the components of systems, and their biological processes, by using the mathematical abstraction of graphs to represent the systems as biological networks [3]. For instance, we can model a protein as a network of amino acids, or an ecosystem as a network of interacting species [4].

Graph theory is the base of this framework. Leonhard Euler, a Swiss mathematician, physicist, astronomer, geographer, logician, and engineer established the first graph theorem in 1736. He was asked to find the best path across the seven Köningsbergs bridges. He solved it by crossing over each of the bridges exactly once [5].

Graphs (G), or networks, offer a natural way to represent systems as a collection of elements that are called nodes or vertices (V), with links between them called edges (E). Following G = (V, E). Graphs (G) and network structures are terms often used interchangeably. Undirected graphs, as shown in Figure 1.1, are composed of an unordered

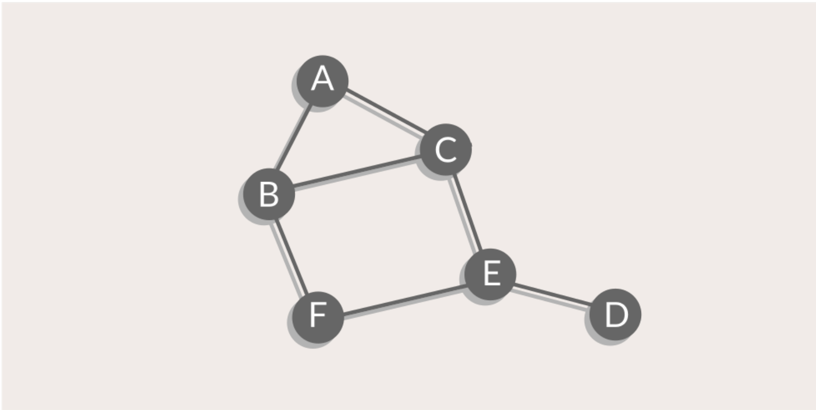

Figure 1.1: An undirected binary graph with six nodes and seven edges.

pair of nodes having interconnections among them. By contrast, a graph with an ordered pair of nodes, with a directional bias from one vertex to another, is called a directed graph. Graphs can be binary or weighted. The edges of binary graphs take the values of either 1 or 0 showing the presence or absence of a connection. Whereas in weighted graphs we describe the edges by weights representing the strength of the link between the two nodes or other quantitative attributes [6].

Many systems take the form of networks: social networks, the internet, and also biological networks. Graph modeling has become an elemental tool for studying the structure, dynamics, and complex behavior of living systems. In a specific manner, the analysis of biological networks in human health has led to an emerging field of research called network medicine. This upsurging discipline is focused on using network topology towards identifying, preventing, and treating diseases [7]. Nodes in a biological network usually represent biological entities of interest, such as genes, proteins, individuals, or species. Edges show interactions between nodes, such as regulatory interaction, information flows, social interactions, or infectious contacts. The representation of the relationships between biological components opens up a wide variety of methods for discovering the mechanistic and functional properties of biological systems, especially in the field of precision oncology [8], enabling the identification and prioritization of disease-causing candidate genes, pathways, and identify possible targets for new drug development, as well as new uses for existing drugs [9]. In particular, in the area of rare diseases, the systems biology approach has helped accelerate the research and lead to a better understanding of the complex disease mechanisms and prediction of the therapeutic responses [10, 11].

Nonetheless, real-world problems are complex and rarely depend on one type of interaction. Thus, the classical single-layer network representation, where networks are isolated and we assume nodes are only connected by a single type of link, as considered up to now, can be an oversimplification [12]. Ignoring the presence of multiple types of links seems to fail against the present heterogeneity of the objects and relations. Therefore, if we want to progress deeper in the study of networks, we need to out-step traditional graphs into a richer scheme capable of characterizing different types of interactions, specifically multilayer networks.

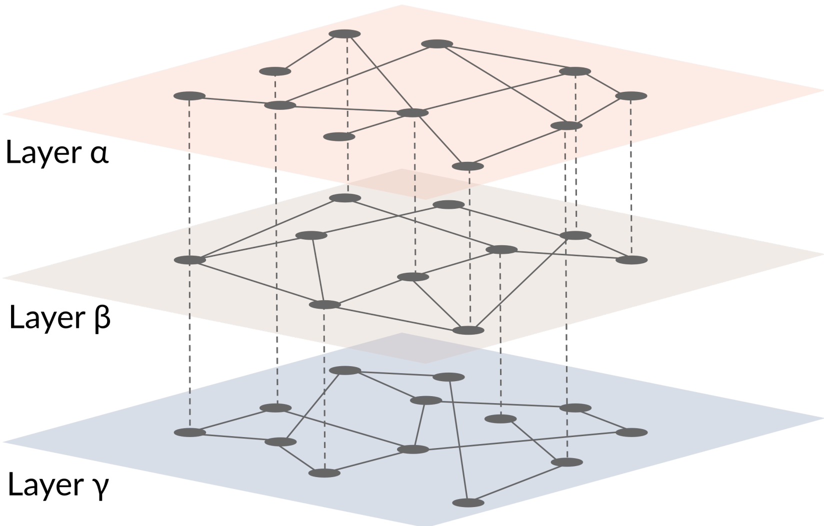

Figure 1.2: A multilayer network comprising three layers: α, β, and γ, with dashed lines representing inter-layer connections, and solid lines representing intra-layer connections.

Multilayer networks are composed of several layers that display interconnections between nodes and their counterparts in another layer, Figure 1.2. Layers represent aspects or features that portray the nodes that belong to that layer. By integrating different layers, one can understand unexpected emergent patterns and potential interactions that contribute, with reference to network medicine, to a disease condition or process. These interactions may otherwise not be uncovered based on a single type of data [13]. Multilayer and multiplex networks are generally used as interchangeable terms, however, the term multiplex describes a particular system of multilayer networks, comprising nodes that represent the same entities in all layers with edges that are restricted to connecting nodes in the same layer [14]. An example of a biological multiplex network could be genes or gene products representing nodes, and the different layers consisting of a protein-protein interactions (PPI) layer, a gene-regulatory layer, and a metabolite layer [15].

The analysis and comparison of networks are fulfilled over the use of different topological metrics that describe the structure and interplay of graph edges while providing a natural way to uncover their prominent features. The most commonly used metrics include degree, path length, closeness and betweenness centrality, and modularity. Specific metrics for the analysis of multilayer networks include the multiplex participation coefficient and the total overlapping degree [15].

Community structure in multilayer networks

An essential structural property of networks is the definition of their community structure, attributable to the interest in finding groups of nodes that share common properties. A community is defined as a set of nodes that are more densely connected among themselves than they are with nodes that are not in the same set [13], as shown in Figure 1.3. Disciplines where systems are often represented as large-scale networks usually display examples of community structures present in them with important applications that prompt research in the field of community detection. For example, biological networks have been observed to be vastly modular [16] with highly connected groups of genes involved in similar biological functions. This plays a crucial role in network medicine, given that disease genes are not scattered randomly in a network, but instead, agglomerate in disease-specific communities, known as disease modules [17]. This is used, for example, to identify the minimal set of genes that characterize disease groups [18].

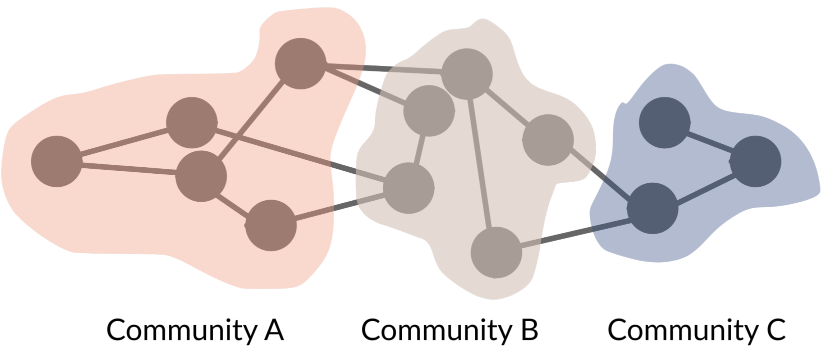

Figure 1.3: Representation of disjoint communities.

Due to its wide applicability and relevance in many fields, the search for community detection strategies, especially for the understanding of complex networks, has been extensively studied. There are several methods and heuristics described, including the Louvain algorithm, a greedy method based on modularity optimization, which tries to maximize the difference between the actual number of edges in a community and the expected number of edges in the community, to extract the community structure of extensive networks. When comparing it to other approaches, it has been shown to outperform the others in terms of computation time, upon which it can be stated that this method has a higher efficiency, and, in addition, a higher quality of the detected communities [19].

Most implementations of the Louvain algorithm have been anchored in the context of single-layer networks, as detecting communities in multilayer networks is computationally much more challenging. Nevertheless, recently, this method has been adapted to multilayer networks [20, 21], by using a generalized concept of modularity and an adaptation of the Louvain randomization algorithm, allowing the analysis of either single-layer or multiplex networks.

Diffusion processes in multilayer networks

Another generalized interest within the area of network theory is understanding the dynamic behavior of spreading processes on networks. This ubiquitous behavior has long been studied in the context of networks, aiming to describe the movement of a continuous quantity in a graph of interacting nodes. The two main diffusion processes that are studied in multilayer networks are epidemic spreading and information diffusion [13]. The concept of network propagation is commonly implemented by a mathematically straightforward algorithm: random walk with restart (RWR) [22]. This iterative and stochastic formulation is based on the assumption that nodes related to similar structural properties tend to settle close to each other. This state-of-the-art algorithm in computational and mathematical analysis allows executing a propagation process over a network, providing a proximity score between two nodes in the graph, as illustrated in Figure 1.4.

Within network medicine, the study of propagation is an active research domain [23, 24], and the spread of disease-associated perturbations within molecular networks has previously been shown for rare diseases [25]. One of the aims of these methods is the identification of the most influential spreaders, as it has been seen that the best spreaders are not necessarily the most connected nodes in the networks [26]. However, most advances in this field perform random walks in a single-layer network. By doing so, they fail to consider the heterogeneous and rich variety of information on networks as well as the fact that the nodes can also spread from one layer to another, and therefore, the dynamical processes on multilayer networks [27]. In view of the increasing interest to generalize this resource to multilayer networks, some methods have developed extensions of the RWR algorithm to explore multiplex networks [28].

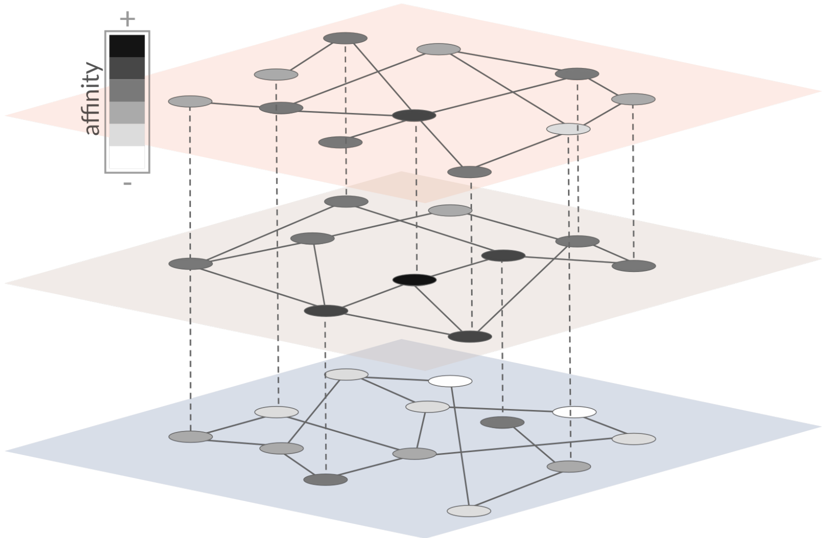

Figure 1.4: The diffusion process is represented on a multilayer network. In this example, the process is seeded with the darkest node in the intermediate layer and the spreading of the diffusion can be appreciated by the progressive fading out of the grayscale colors of the other nodes in all layers.

1.2 Case study: Hepatoblastoma

Childhood cancer is rare, with only 35,000 new cases each year in Europe [29]. Even though advances in diagnosis and care have yielded a significant improvement in survival rates over the last decades, cancer remains the leading disease-cause of death in children in developed countries [30]. The most common types of cancer diagnosed in children 0 to 14 years of age are leukemias, brain cancer, lymphomas, and solid tumors, such as neuroblastoma and Wills tumors.

Fundamentally, cancers in children are biologically different from adults. Showing the fewest somatic mutations prevalence across human cancer types, hence having reduced molecular target options [31]. Despite this substantial difference with adult cancer, the low incidence of pediatric cancer prevails and these patients are faced with challenges, such as late or incorrect diagnosis, difficulties finding clinical expertise and accessing appropriate treatments, high uncertainty in clinical decision-making, scarcity of available registries, and tissue banks, insufficient medical training on each specific disease area, lack of information among the general public, and apparent difficulty to carry out clinical studies due to the small number of patients. Which has disincentivized the biopharmaceutical industry from investing in pediatric research to develop new therapies [32].

In 2019, the European Commission funded a research consortium named Individualized paediatric Cure (iPC) (ref. 826121), focused on identifying effective personalized medicine for pediatric cancers. This project comprises 21 partners, including the Barcelona Supercomputing Center (BSC) and the Germans Trias i Pujol Research Institute (IGTP), from eleven different countries. The goal of the iPC project (https://ipc-project.eu/) is to achieve an effective personalized medicine for pediatric cancers by collecting, producing, standardizing, and harmonizing existing clinical knowledge and medical data and, with the help of artificial intelligence, creating treatment models to predict optimal standards and experimental therapies for each child, thus improving both their survival and quality of life.

One target of this multidisciplinary project is to improve the clinical treatment of hepatoblastoma (HB), a very rare type of liver cancer, which is most frequently diagnosed in children less than 5 years of age, and, it represents approximately 1% of all cancers in this age group with an incidence of 1.8 cases per million children [33]. HB is a tumor that originates in the liver and rarely metastasizes to other areas of the body, symptoms include a tumor mass, bloating, and abdominal pain, and the diagnosis is made with blood tests, imaging tests, and a biopsy [34]. The treatment of HB has been one of the great success stories in childhood cancers, combining efficient chemotherapy and surgical approaches that led to dramatic improvements in outcomes for HB patients, with survival rates increasing from 20 to 80% over the past decades [35]. However, in clinically advanced or aggressive tumors the choice of treatments is limited, with event-free survival in children, up to 3 years of age, of 34% [36]. In addition, patients who survive may suffer serious and lifelong side effects from chemotherapy and immunosuppression.

Even though it is the predominant liver cancer in children and incidences of HB are increasing as much as 2 to 3% per year [37], clinical and molecular investigations of the disease are limited and effective treatment options are lacking, in some measure due to the difficulty of collecting and analyzing a large series of samples, caused by its low incidence. The first case of HB was described in the English literature by Willis, as an embryonic tumor that contains hepatic epithelial parenchyma [38]. At that time, HB was not distinguished from hepatocellular carcinoma (HCC). Consensus over the classification of pediatric liver tumors, histological subtypes, and molecular stratification of HB, has been a constant topic of study over the years.

Studies have shown that HB is genetically a very simple cancer, compared to hepatocellular carcinoma, liver cancer in adults, with an average of only 3.4 mutations per tumor, which is the lowest mutation rate ever described for human cancer. The most common alterations in HB, with an incidence of around 70%, are activating mutations of the β-catenin encoding gene, catenin beta 1 (CTNNB1) [39]. Exome sequencing studies have revealed NFE2L2 as the second most mutated gene in approximately 10% of cases [40]. Thanks to all these findings, high-throughput technologies allowed the molecular subtypes identification of various cancers and their associated oncogenic aberrations. Two subclasses of HB, C1, and C2, have been identified, which resemble the late and early stages of liver development [41], and recent studies described a third subclass of HB, named C2B [42].

Chapter 2

Aims of the study

This work involves the collaboration between two lines of research, the multilayer networks at the Computational Biology Group at Barcelona Supercomputing Center (BSC) and the HB-focused line at the Childhood Liver Oncology Group at Institut d'Investigació en Ciències de la Salut Germans Trias i Pujol (IGTP). This project is based on the hypothesis that by determining how gene dysregulation propagates on a multilayer network, we can uncover novel candidate cancer genes for rare diseases. The objectives are:

1. To characterize the landscape of high-throughput transcriptomics data in HB and identify genes dysregulated through a differential meta-analysis, from the most relevant studies.

2. To discover and analyze the multilayer community structure of HB.

3. To determine how gene dysregulation propagates on a multilayer network and identify the genes that are most influential in the dysregulation spreading of expression dysregulation.

4. To develop a tool that encompasses these analyses and deploy the workflow to the Virtual Research Environment of the iPC project (iPC-VRE).

Chapter 3

Materials and methods

3.1 MetaDEGs of HB

Gene expression profile data

We selected five RNA-Seq studies as a representation of the landscape of high-throughput transcriptomic data of hepatoblastoma (HB): GSE133039, GSE89775, GSE104766, GSE151347, and GSE81928, following the expertise of the IGTP Children's Liver Oncology group, led by Dr. Carolina Armengol, co-director of this work. All datasets were available in the Gene Expression Omnibus (GEO) database (www.ncbi.nlm.nih.gov/geo). Detailed information on these datasets is shown in Table 3.1. We downloaded the raw counts table and metadata from the GREIN Interactive Navigator [43] (www.ilincs.org/apps/grein). The larger cohort is composed of 64 samples, with 32 cases and the same amount of controls, by contrast, the smaller study spans over 10 case samples and 3 controls. We analyzed in total 181 samples from 101 patients. This approach is still more important specifically because of the rarity of this disease since it allows the study of a large series of samples.

Table 3.1: Characteristics of the five RNA-Seq datasets in our integrated analysis. PMID: PubMed Identifier.

AUTHOR	Year	GEO dataset	PMID	Cases (n)	Controls (n)
Carrillo-Reixach	2020	GSE133039	32240714	32	32
Ranganathan	2016	GSE89775	27910913	10	3
Hooks	2018	GSE104766	9152775	25	25
Wagner	2020	GSE151347	32824198	11	11
Valanejad	2018	GSE81928	30271949	23	9

Dataset preprocessing and screening for DEGs

Using the DESeq2 R package [44] we normalized each dataset by the median of ratios function, and we identified the differentially expressed genes (DEGs) between cases and control samples. Genes with a corrected p-value, given False Discovery Rate (FDR) < 0.05 and $|\log2 \text{ fold change (FC)}| > 1$ were considered DEGs. The up- and down-regulated gene lists were sorted by log2FC for each dataset and were saved for subsequent integration analysis.

RRA integrated analysis

We used the robust rank aggregation (RRA) methodto integrate the five RNA-Seq datasets, which is a standard method that can minimize bias and errors between multiple datasets. By adding this step to the workflow we can detect genes, referred to as metaDEGs, whose ranking is consistently better than expected under the null hypothesis of uncorrelated inputs and determine a significance score for each one of them. The lists of log2 fold change (log2FC or LFC) values from each study were merged to get a combined matrix, and we applied the RRA method for the integration with the RobustRankAggreg R package [45]. This package has been designed to integrate results from different platforms and procedures, including RNA-Seq and microarray, without the need of having all genes present in the different studies. Genes with a significant RRA score < 0.05 were considered as robust DEGs. Furthermore, we used the pheatmap R package to draw a clustered visualization of the top dysregulated genes.

Between-study heterogeneity

Following the combining of the different expression studies through the meta-analysis, we wanted to ensure that the inter-study heterogeneity between the datasets was minimal. We used the ggridges Python package to plot the overlapping line plots of each distribution, aiming to visualize the changes present. To estimate the divergence between the distribution functions of the studies and determine that these are hence homogeneous, we used the Wasserstein metric, also known as Earth mover's distance, which is a nonparametric method employed within the waddR R package [46]. This method tests whether two distributions FA and FB are different, by testing the null hypothesis H0: $FA = FB$ against the alternative H1: $FA \neq FB$. To determine if the paired studies do not show a significant differential distribution, we set the cutoff criteria to p-value $<$ 0.05.

Table 3.2: Characteristics of the three layers in our multilayer network study.

Relation Type	Source	Number of nodes	Number of edges
Metabolic reactions	BiGG Models (UCSD)	1786	52077
PPIs in the liver	IDD (UToronto)	11729	960872
Pathways	Reactome (ELIXIR)	10684	885479

3.2 Community structure

Assembling a multilayer network

We constructed a multilayer network consisting of three unweighted and undirected layers, obtained from different publicly available sources. In this multilayer network, nodes represent genes and edges represent different types of associations, including metabolic reactions, protein-protein interactions (PPI) in the liver, and pathways. First, we retrieved the metabolic reaction associations between human genes from Recon3D [47] in the BiGG Models - UCSD database. In this layer, nodes are connected if they are involved in the same metabolic reactions. As a result, we sourced 1,786 genes and 52,077 relations from this model. Second, we constructed the PPI network from associations specific to the human liver, from the Integrated Interactions Database (IDD) - UToronto [48]. In this layer, the edges represent the physical connections between the gene products, and it is composed of 18,726 nodes and 960,872 edges. Finally, we built the pathways network from the Reactome [49] database, a curated and peer-reviewed source that defines a relationship between two genes if they are annotated in the same pathways. As a result, we obtained 10,682 nodes and 885,489 edges. Summarized statistics of each network are shown in Table 3.2.

To further analyze and understand these networks, we used one of the state-of-the-art Python packages for the study of their structure, NetworkX, which allowed us to calculate descriptive metrics such as the diameter of the graphs, the network density, and the degree of the nodes. We define the diameter of a network as the shortest path between the two most distant nodes. The network density is defined as the ratio of observed edges to the number of possible edges for a network. The degree of a node is defined as the number of connections that it has with other nodes in the network [50]. In addition to the aforementioned graph analysis, we used an interactive visualization tool, Graphia, available at www.graphia.app [51], to explore and better comprehend each of the three layers independently.

Finally, we employed the matplotlib_venn Python package to draw a Venn diagram to represent the intersection of nodes between the layers.

Community Detection

We used MolTI-DREAM [21], an implementation developed within Disease Module Identification DREAM Challenge, to define the community structure of the multiplex network. This approach uses an adapted version of the Louvain algorithm to optimize the modularity and therefore, define the community structure of the multilayer network. The multilayer modularity measure to detect communities in multiplex networks can be written as

$$\sum_{g} \frac{w^g}{2m^g} \sum_{i,j} (Z^g_{i,j} - \gamma \frac{s_i^{(g)} s_j^{(g)}}{2m^g}) \delta_{c_i,c_j}$$

[21] where X^g defines the network of the layer g, w^g is the weight associated with the network g, m^g is the sum of the weights of all the edges of that layer, $X^g_{i,j}$ is the sum of weights of the edge i,j, when i j, S^g_i is the sum of the weights of all the edges involving vertex i in X^g, γ is the resolution parameter modulating the size of the communities detected, and δ_{c_i,c_j} is equal to 1 if i and j belong to the same community and to 0 otherwise.

In comparison to previous approaches [20], the employed version of this software incorporates the randomization procedure, which improves the accuracy of the detected communities, in particular for dense multiplex networks [21]. As the different number of randomizations used by the developers (5, 10, and 15) yielded a similar amount of communities in their results, we used the lower amount to improve the time efficiency of the workflow. We filtered the resulting communities to only keep the significant clusters containing a minimum of 7 nodes, as recommended by the authors [21]. Eventually, to identify the optimal value of the Newman modularity resolution parameter (γ) for the study, we tested different γ to find the value where the average community size, as a function of the number of communities, establishes a plateau, and, therefore, mark the limit of the γ values range to be considered. Modularity can be described as a scalar value that measures the relative density of edges inside communities compared to edges outside communities. Optimizing the value hypothetically results in the best possible partition of nodes of a given graph.

3.3 Gene dysregulation diffusion

Affinity Scores

To determine the closeness and relations of the network nodes, we computed the affinity scores (AS) between each DEG, used as seed nodes, and all other nodes using the random walk with restart algorithm (RWR). This method calculates the level of affinity between two nodes by letting a random walker, starting at the seed node, iteratively move to a neighbor node, and restart from the seed node with a defined restarting probability, in this study set to 0.15. We used the Python package multiXrank [28] to perform this analysis as it permits the exploration of multilayer networks. Once all nodes were traversed, we got a probability vector containing the AS of all nodes for each metaDEG, and subsequently, we constructed an AS matrix by scaling the features, in the range $\in$ [0,1], and joining the resulting vectors.

AS-DE correlation analysis

To estimate the spreading of dysregulation in a multilayer network within each community, we calculated if the affinity score (AS) and the differential expression (DE) values were correlated, and we ranked the results to determine which genes had a more relevant role in this process. First, we appended a vector with the DE values present in the RNA-Seq datasets to the AS matrix. Then, recursively, we subset the matrix to represent the genes in each of the defined communities, and we computed the Spearman's rank correlation coefficient (ρ) between the DE and the AS of each gene in the community. The Spearman's ρ measures both the strength and the direction of the relationship between the ranks of data and is defined as

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

[52] where ρ defines the Spearman's rank correlation coefficient, d_i is the difference between the two ranks of each observation, and n is the number of observations. It can result in any value from -1 to 1, and the closer to the absolute value of the coefficient to 1, the stronger the relationship.

We used the spearmanr functionality from the scipy Python package to perform this nonparametric statistic. For each metaDEG, we obtained a correlation value and the corresponding p-value, which in this case, defined the probability of seeing the observed correlation or stronger if no correlation exists. We used this p-value to filter out the metaDEGs with a weak significance with a cut-off value of 0.05, and as a result, we obtained a list with all metaDEGs with a significance Spearman's ρ, that we ranked to define the genes with a higher correlation between the differential expression and the affinity score. To interpret the results, we followed the general guidelines, and divided the absolute values into three strengths of association: weak (below 0.29), moderate (between 0.3 and 0.69), and strong (between 0.7 and 1).

Structural measures for multiplex networks

To better understand and characterize the multiplexity of the system, particularly of the genes with a significant AS-DE correlation, we used two metrics: the participation coefficient and the total overlapping degree. Node degree accounts for the total number of connections of a node, and when working within the context of multilayer networks, we used the total overlapping degree, o_i, to describe the total number of nodes in the intersection of the layers. Defining the edge overlap of nodes i and j as o_{ij}, we introduce the overlapping degree of node i as,

$$o_i = \sum_j o_{ij} = \sum_\alpha K_i^{[\alpha]}$$

[53] with $o_i \geq$ ki. Conversely, we define the multiplex participation coefficient P_i of node i,

$$P_i = \frac{M}{M-1}[1 - \sum_{\alpha=1}^{M}(\frac{K_i^{[\alpha]}}{o_i})]$$

[54], where the multiplex participation coefficient, P_i, takes into account the heterogeneity in the number of links of the node across the different layer, returning values $\in$ [0, 1], measuring whether the connections of node i are primarily concentrated in just one layer (coefficient P_i would be equal to 0), or are rather found uniformly distributed among the M layers, with the same number of edges in each of them (Pi equivalent to 1) [15].

Pathway enrichment analysis

The R package gprofiler [55] was used to annotate the genes from the highest correlated genes in the previous analysis, along with the other genes from their community. This resource performs statistical enrichment analysis to find an over-representation of gene functions from biological ontologies such as Gene Ontology (GO) and molecular pathways such as KEGG and Reactome, human disease annotations. The significance of the over-representation is calculated through the hypergeometric test followed by a correction of multiple tests, given $FDR < 0.05$. In our case, we used gprofiler to annotate solely the pathways, by using all Homo sapiens annotated genes present in the KEGG [56], Reactome [57] and Wikipathways [58] databases, and correcting all p-values by setting a cut-off of 0.05, in addition to the default parameters. To visualize the results we used the ggplot2 R package.

3.4 Implementation of the multiAffinity tool

Processing the workflow

To transform the described analysis into an automatized workflow, we unified each step using Command-Line (CL) scripting. CL, also called shell, which is prebuilt into macOS and Unix systems, and available on Windows through intermediate tools, and can be described as the primary program that terminals use to receive commands and return the produced information, allowing the user to efficiently perform many tasks. We employed an implementation of the shell system, Bash, also known as the "Bourne Again SHell", as a reference to the Bourne shell, which it replaced in 1989 [59]. Many computational disciplines, such as bioinformatics, rely heavily on the CL to do tasks such as wrangling files, handling big data, manipulating spreadsheets, and parallelizing and automatizing their work [60].

We curated the shell commands in the subsequent scripts, creating a tool named multiAffinity. We shared it in a version-controlled Git repository, Github (www.github.com), and presented a containerized image of the package on the Docker (www.docker.com) and Github repositories. This step was followed to achieve the highest degree of reproducibility for easy and simple usage of the workflow with Linux, Windows or MacOS operating systems.

Deployment to the iPC-VRE

The iPC Virtual Research Environment (VRE) is a cloud-based infra-structure being developed by the Spanish National Bioinformatics Institute (INB) / ELIXIR-ES node, in the Barcelona Supercomputing Center (BSC), within the context of the Individualized Paediatric Cure (iPC) project. This user-friendly interface brings together data and tools to facilitate the research of other groups in the consortium. The user can analyze the sensible and private data hosted within the catalog portal, or upload their own data and run the deployed tool. Implementing multiAffinity to the iPC-VRE will simplify its access to a wide community of cancer researchers and clinicians without compromising their confidentiality.

To deploy multiAffinity to the iPC-VRE we had to transform the bash scripts into a more standardized form using the Common Workflow Language (CWL), an open-source specification for designing portable and scalable workflows. The desired CWL workflow consists of three main pieces: a .yaml file specifying the inputs of the workflow; a .cwl file containing the workflow, describing how the different components should be executed and what the outputs are expected to be; and a set of .cwl files specifying the constraints about the ways the tools will run.

Chapter 4

Results

4.1 MetaDEGs of HB

Screening for DEGs

In this study we performed a multi-step bioinformatic analysis to reveal the key genes of HB involved in the spreading of dysregulation in a multilayer network. The flow diagram employed is shown in Figure 4.1.

To obtain the metaDEGs of HB, first, the DEGs comparing HB and normal liver tissue samples were obtained from GSE133039, GSE89775, GSE104766, GSE151347, and GSE81928. The distribution of the DEGs in each dataset is shown in Figure 4.1a. The analysis of the GSE133039 study identified 2970 up-regulated and 1710 down-regulated genes, GSE89775 included 122 up-regulated and 5 down-regulated genes, the GSE104766 dataset defined 611 up-regulated and 982 down-regulated, the GSE151347 defined 1398 up-regulated, and 1158 down-regulated genes, and finally, the GSE81928 dataset identified 174 up-regulated and 15 down-regulated genes.

RRA integrated analysis

With the robust rank aggregation (RRA) method, applied with the RobustRankAggreg package, we could identify 323 genes differentially expressed across the 5 independent studies, with 199 up-regulated and 124 down-regulated genes. The top 20 most significant metaDEGs, up-regulated and down-regulated respectively, are shown in Figure 4.2b.

Between-study heterogeneity

Following the meta-analysis, we assessed the heterogeneity of the five different utions to determine that the inter-study batch effect is not significant and would not affect the

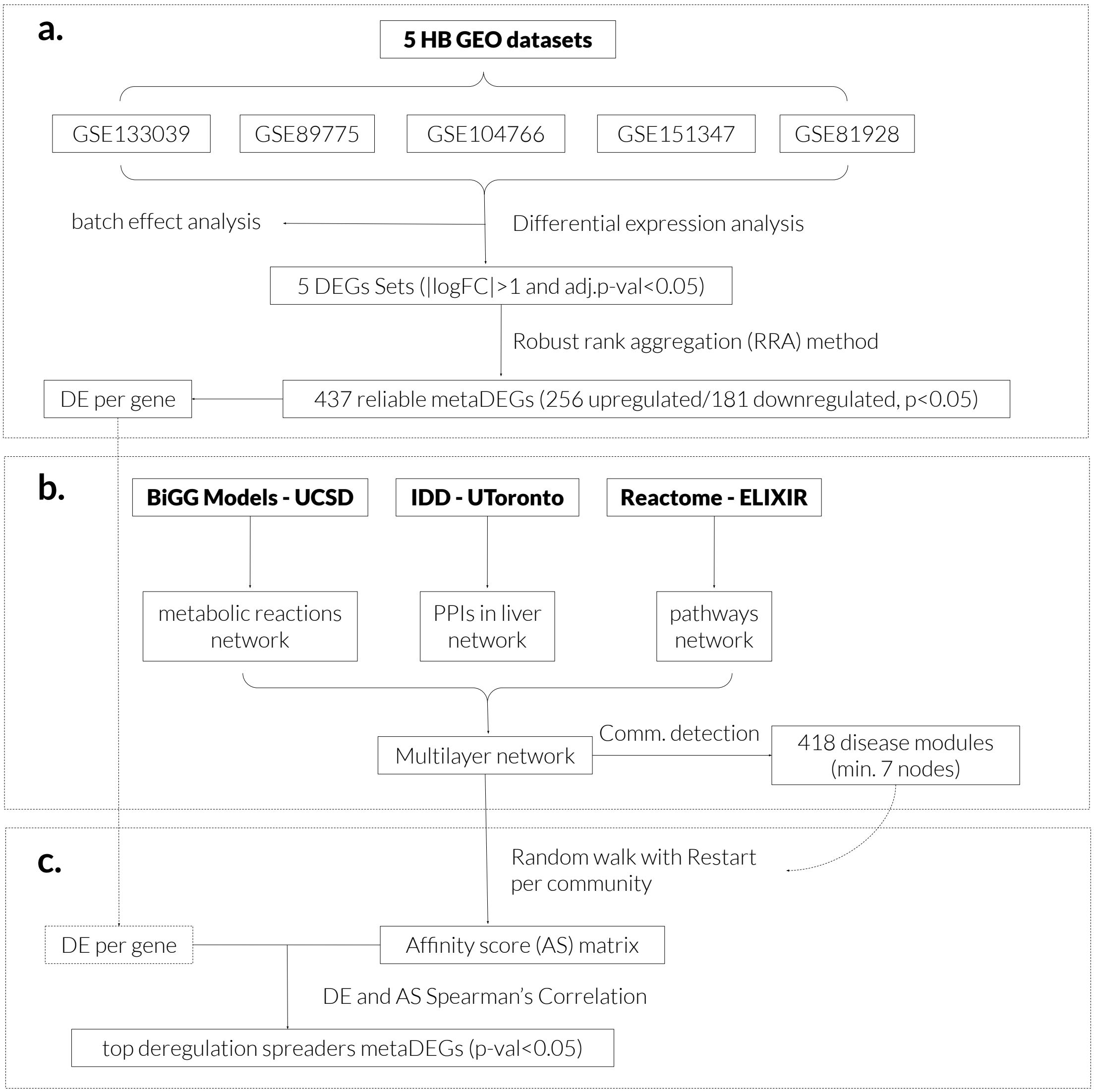

Figure 4.1: Study design of the bionformatic analysis in the present study. Flowcharts for (a) obtaining metaDEGs of HB, (b) defining the community structure in the multilayer network, and (c) determining the AS-DE correlation to rank the top genes involved in the spreading of dysregulation.

results, Figure 4.3.

We implemented the 2-Wasserstein distance as a metric to detect differences between each pair of distributions. After performing the computation, we could determine that the p-value between the GSE81928 and GSE89775 studies was 0.05, then, GSE81928 and GSE104766 showed a p-value of 0.88, GSE81928 and GSE133039 a p-value of 0.83 and finally, the result between GSE81928 and GSE151347 was 0.85. These probability values are not smaller than the established cutoff criteria for this test, 0.05, and therefore, we cannot affirm that these distributions are statistically different, and therefore we did not remove any study from this analysis. Although out of the scope of this work, we recommend a deeper statistical analysis of any subtle difference among these distri-

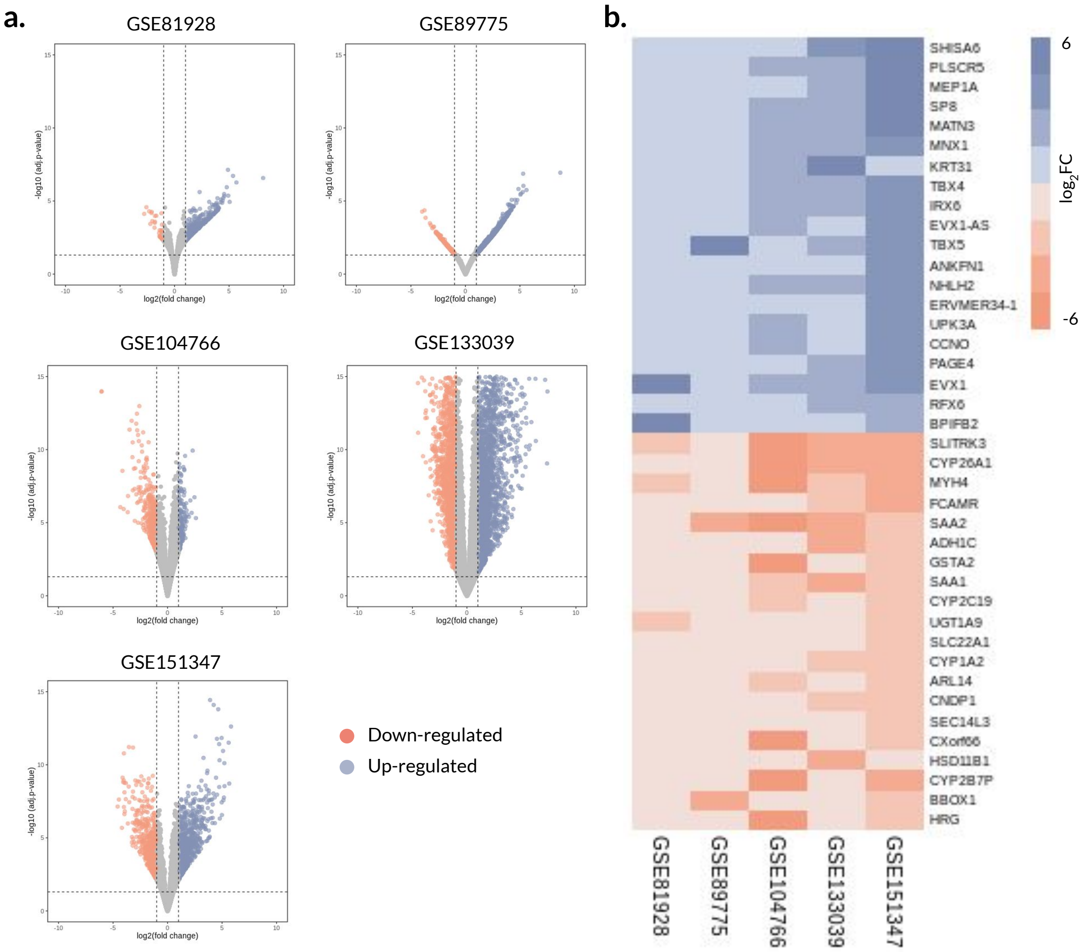

Figure 4.2: Identification of DEGs in HB between tumor and normal tissues using the RNA-Seq datasets from five different studies. (a) Representative volcano plots of the datasets. Blue points represent up-regulated genes, while red dots indicate down-regulated genes. Grey points represent genes with no significant difference. (b) The log2FC heatmap of each RNA-seq dataset for the top significant metaDEGs.

butions and the consideration of batch effect correction guidelines [61] to confirm this result.

4.2 Community structure

Assembling a multilayer network

Before performing the chosen community detection method, we inspected the three gene-to-gene networks described previously: metabolic reactions, PPIs in the liver, and pathways. A representation of each of these layers is shown in Figure 4.4a.

We computed different metric estimations to determine the density, diameter, connectivity, and degree of each layer. First, having in mind that the density values range

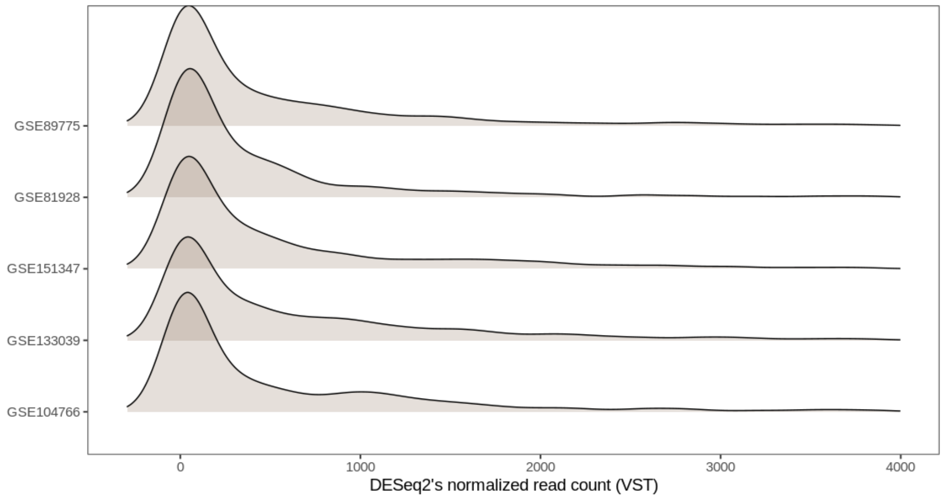

Figure 4.3: Ridge plot showing the distribution of the normalized read count for the different RNA-seq datasets.

from 0 (no link at all) and 1 (all nodes are connected), we found that the graph of the metabolic reaction has the lowest density, with a value of 0.033, followed by the pathways, with 0.016, and the PPIs in liver relations with 0.0084. As to the diameter of the largest components of each network, we can affirm that the greater belongs to the metabolic reactions layer, with 11 nodes, whereas the pathways and PPI in liver layers both have a maximum distance of 7. As for the nodes with the highest number of links attached, therefore the most connected genes, we find the SLC3A2, the TMEM27, and the SLC6A18 genes, with 328, 299, and 289 relationships respectively, in the metabolic reactions layer. The nodes with the highest degree in the layer of liver-specific PPI are YWHAE, APP, and TRIM25, with 3420, 2067, and 1913 links respectively, while those in the pathway layers are RPS27A, UBA52, and UBB genes, with 2505, 2505, and 2342 links respectively.

Regarding the interrelation among the three layers, we computed the sum of the overlapping genes among them. The Venn diagram represented in Figure 4.4b shows the number of genes for each combination. We see that the PPIs in liver and pathways layers share the most amount of nodes, with 5937 nodes, whereas the PPIs in liver and metabolic reactions layers share the least, with only 72 nodes. The three layers partake in a total of 1207 overlapping nodes.

Community Detection

When tuning the resolution parameter (γ) in MolTI-DREAM, the most substantial changes in the number of communities and their size are seen in an initial range of γ,

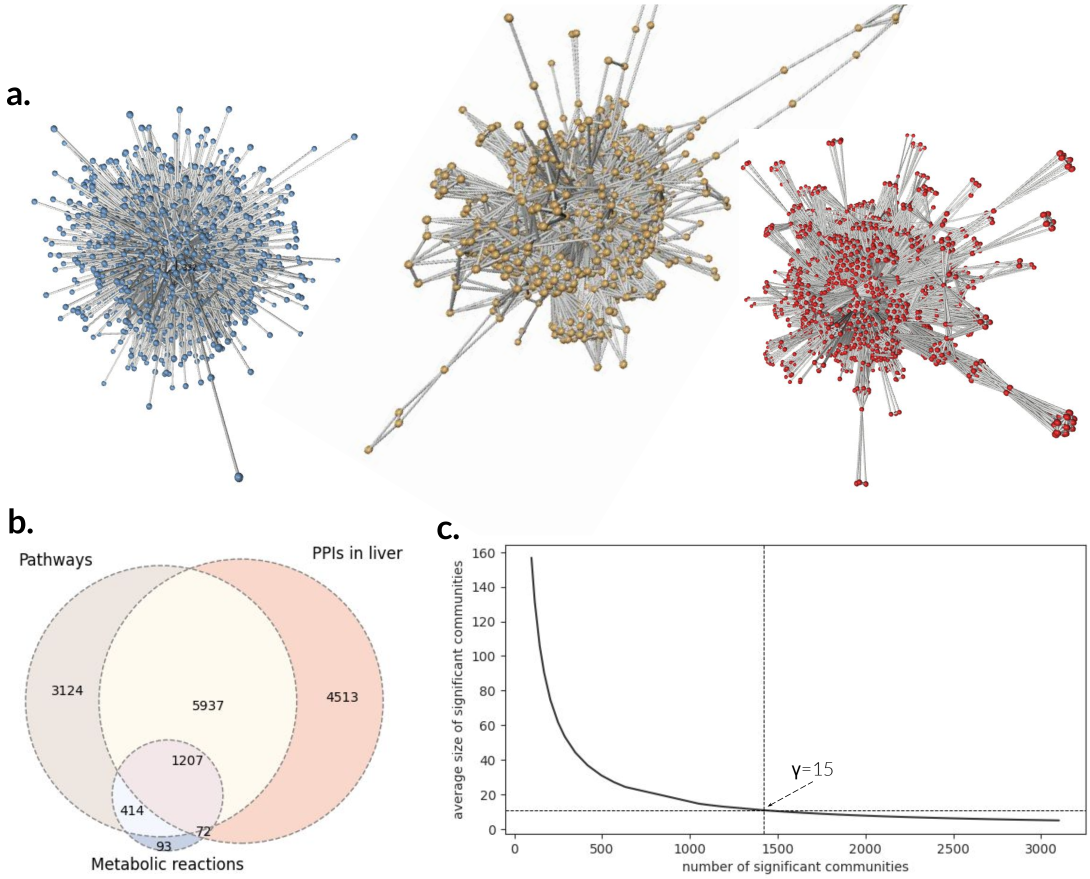

Figure 4.4: Multilayer networkcomposition and Community detection. Network visualization of (a) metabolic reactions in blue, pathways in brown and in red, PPIs in liver relations, using the Graphia package software. (b) The Venn diagram represents the relations between the nodes in the three layers. (c) Identification of the endpoint of the range of resolution values for the study, $\gamma \in$ [0-15].

before the average community size, defined as a function of the number of communities, establishes a plateau, with a 0.001 margin of error. By iterating through different γ values, we established the endpoint of this range (γ=15), where we determined 1420 communities with an average size of 10.82 nodes, as shown in Figure 4.4c.

4.3 Gene dysregulation diffusion

Correlation between AS and DE

The RWR algorithm provides the affinity scores of all nodes in the multilayer network for the seed nodes (each DEG), with a restarting probability of 0.15. The results were grouped to form an affinity score (AS) matrix, as represented in Figure 4.5, showing high scores for the nodes that are closely connected to the seed node and low scores for the others. To complete the analysis, we added a vector denoting the differential expression

(LFC) values for each node to the AS matrix, and we computed the Spearman's rank correlation for each metaDEG previously identified. To obtain the final AS-DE correlation, we discarded the non-significative ρ by using the defined p-value as a threshold, obtaining a ranked list of genes with a higher AS-DE correlation ρ. These genes can be interpreted as those that are more involved in the spreading of dysregulation, following our initial hypothesis.

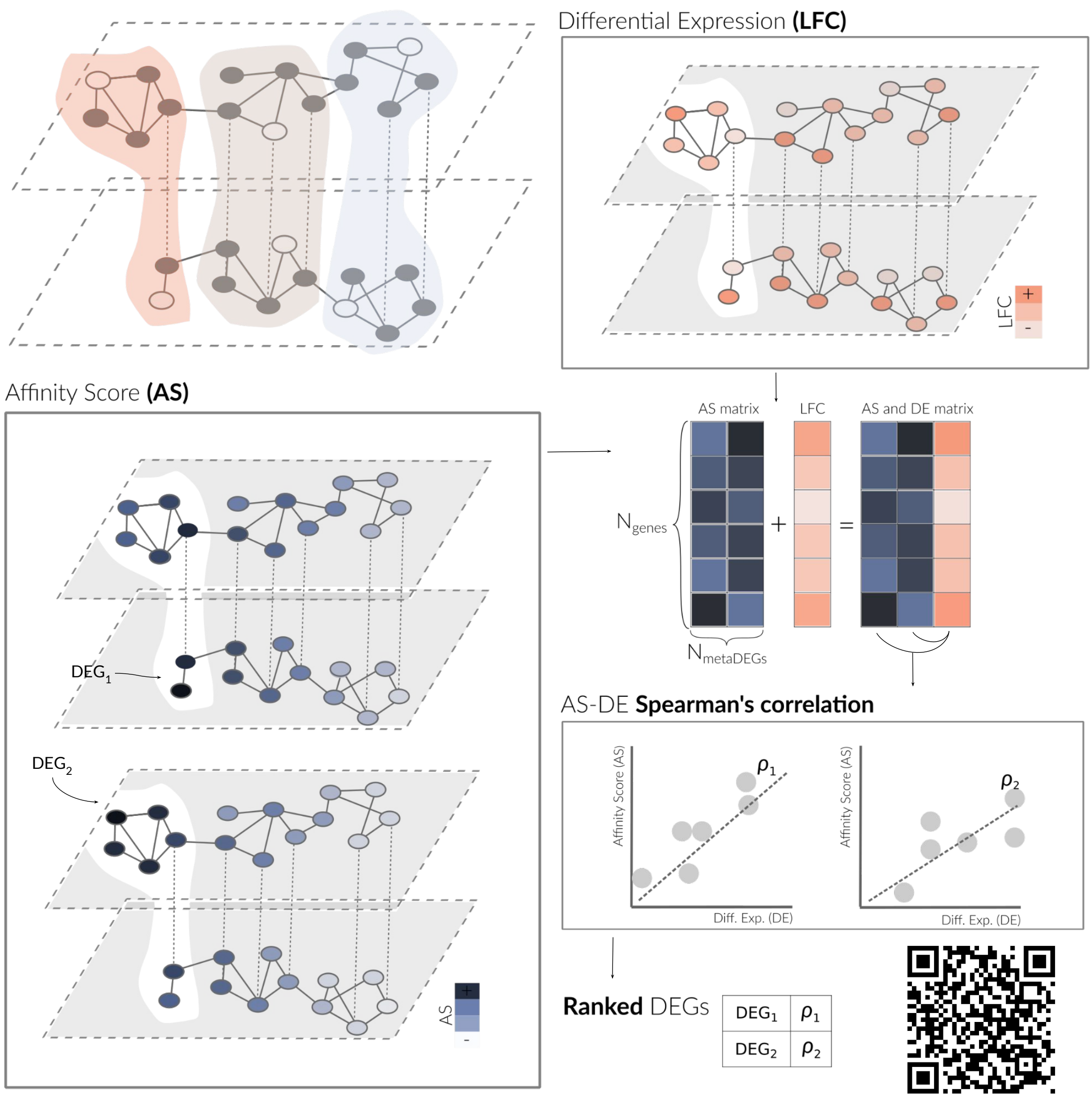

Figure 4.5: Simplified schema of the process to rank the DEGs most involved in gene dysregulation spreading for one community, including a QR code to visualize the schema in video format (https://youtu.be/1tcwczu47aI).

Using the described layers, representing metabolic reactions, PPI in the liver, and pathways, and the described parameters, we obtained different amounts of significantly correlated metaDEGs when iterating through the resolution range of $\gamma \in$ [0-15]. As shown in Figure 4.6, we obtained the larger number of genes when setting the resolution parameter to 8, obtaining a total of 41 candidate genes with a significant ρ, p-value $<$ 0.05, between the corresponding affinity score (AS) and differential expression (DE) absolute values. In this scenario, we defined a total of 418 disease modules, or communities with more than 6 nodes, that had an average size of 36.76 genes.

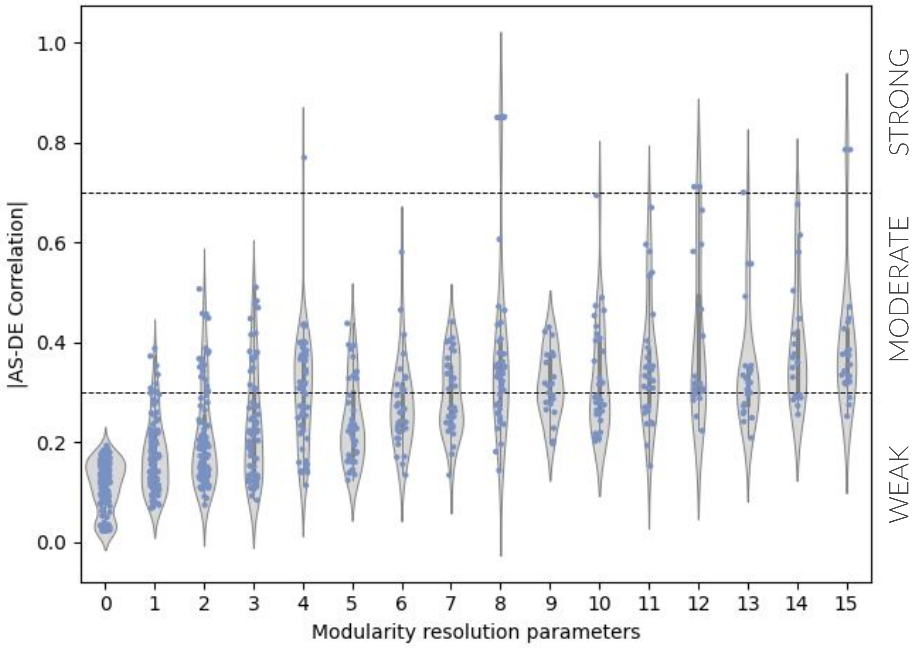

Figure 4.6: Scatter and violin plots comparing the AS-DE correlation results with the defined modularity values range [0-15].

Four genes showed a strong correlation, ρ larger than 0.7: LRRC52, DLX6, DLX5, and MSX2, all with an absolute value rounded of 0.85, as shown in Table 4.1. Interestingly, these 4 genes are not expressed in any liver cell type, reinforcing the idea that they are HB specific [62]. With a moderate strength of association, absolute ρ between 0.3 and 0.69, we found 24 genes, including NAT2, CST4, KCNU1, and HAMP with values of -0.61, -0.47, 0.46, and -0.44. Additionally, we defined 13 genes with a weak Spearman's correlation, with absolute values below 0.29.

LRRC52, as shown in Figure 4.7a, has the highest AS-DE negative correlation when using the community-based (local approach). Whereas if we compute AS-DE correlation using all genes (global approach), we cannot discern a dysregulation effect of LRRC52 on a specific group of genes, as seen in Figure 4.7b. LRRC52 stands for Leucine Rich Repeat Containing 52 and is a protein-coding gene, which has been related to the regulation of the calcium-activated potassium channels (BK) function and location [63]. Expression in embryonic tissues and stem cells has been located in epithelial cells and the liver [64]. An association between this gene and HB is not described in the literature, yet it

Table 4.1: Correlation values, with the corresponding community details and differential expression values for the genes with the highest AS-DE correlations.

metaDEGs	AS-DE Corr (ρ)	Community Size	Community ID	log2FC
LRRC52	-0.85	13	368	4.52
DLX6	0.85	9	252	3.74
DLX5	0.85	9	252	1.52
MSX2	0.85	9	252	4.29
NAT2	-0.61	30	146	-1.48
CST4	-0.47	28	146	6.28
KCNU1	0.46	36	272	3.88
HAMP	-0.44	74	114	-1.31
TDO2	-0.44	31	146	-1.20

has been described as an important gene connected with Lymph Node Carcinoma [65]. An important paralog of this gene is LRIG1, which is a positive prognostic marker in hepatocellular carcinoma, the main primary liver cancer in adults [66]. The 13 genes belonging to the community of LRRC52 upon which this gene may exert dysregulation effects (Community ID 368) are ADPRH, ANKRD35, ARGLU1, ARL6IP4, CCDC28A. HP1BP3, NCBP3, NOP16, NSG1, OCEL1, RSRC2, SLC25A36, and YPEL2.

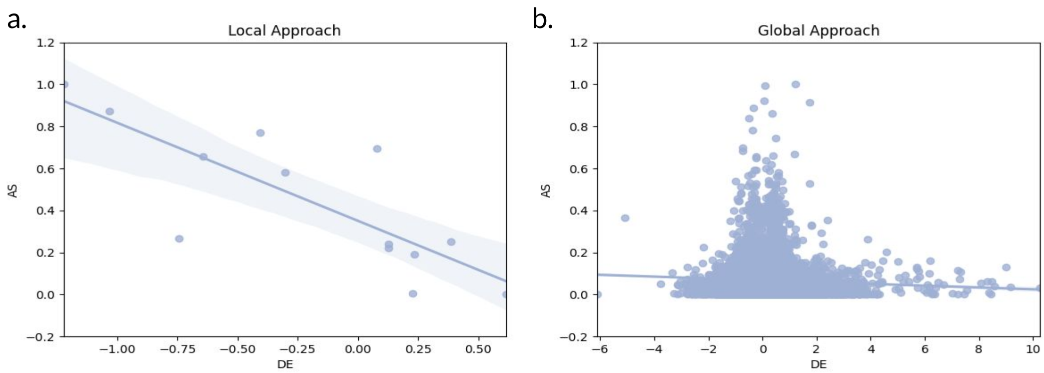

Figure 4.7: Scatter and violin plots comparing the AS-DE correlation results with the defined modularity values range [0-15].

The second and third genes with the highest ρ; DLX6 and DLX5, are protein-coding genes that encode for members of a homeobox transcription factor gene family, which current research holds to be important in appendage development. DLX5 and DLX6 can be seen to work in conjunction and are both necessary for proper craniofacial, axial, and appendicular skeleton development [67]. Mutations in the DLX6 and DLX5 genes have been shown to be involved in the hand and foot malformation syndrome [68]. Hypermethylation of DLX6-AS1 has been identified as the most significant classifier of different molecular HB groups, also associated with poor predicted prognosis [69].

Increased expression of DLX5 has been described as a determinant for the proliferation of lung and ovarian cancer cells [70]. In several mouse models of T-cell lymphoma, it has been shown to act as an oncogene by cooperating with activated Akt, Notch1/3, and Wnt to drive tumor formation [71].

MSX2 encodes a member of the muscle segment homeobox gene family, which is a transcriptional repressor, with previous observations implying that its normal activity may establish a balance between survival and apoptosis of neural crest-derived cells required for proper craniofacial morphogenesis [72]. The encoded protein, which acts as a transcriptional regulator in bone development, may also have a role in promoting cell growth under certain conditions and may be an important target for the RAS signaling pathways [73]. Related to the other described genes, MSX2 is known to antagonize the stimulatory effect of DLX5 on ALPL expression during osteoblast differentiation. Diseases associated with MSX2 include Craniosynostosis 2 and Parietal Foramina With Cleidocranial Dysplasia. Among its related pathways are Mesenchymal Stem Cells and Lineage-specific Markers and Neural Crest Differentiation. It acts as a transcriptional regulator in bone development, represses the ALPL promoter activity, and antagonizes the stimulatory effect of DLX5 on ALPL (alkaline phosphatase) expression during osteoblast differentiation [74]. MSX2 expression is induced by Wnt signaling pathway activation and is a direct β-catenin/TCF target [75].

DLX5, DLX6, and MSX2 belong to the same community (Community ID 252) and results suggest that they may influence the dysregulation of 9 other genes, which are ABCC11, CCDC92, HINT3, OAZ3, PSMD4, PSMD8, SAT2, SIL1, and TMEM242.

Structural measures for multiplex network

To understand the roles of the candidate genes in the multilayer network, we studied how the degree is distributed among the different nodes at each layer and how it is distributed across different layers, by using two metrics: total overlapping degree and the multiplex participation coefficient. The first measure, o_i, takes into account the total number of nodes in the intersection of the layers, and the second, P_i, describes whether the connections of each node are uniformly distributed among the different layers, the larger the value, the more equally distributed is the participation of this node.

By defining the z-score of the total overlapping degree and the participation coefficient throughout the significant DEGs we can classify the nodes between "hubs" and "leaves", if they have many or few connections, and "focused", "mixed", and "multiplexed", if those connections are distributed on one or more layers, respectively. As shown in Figure 4.8a none of the top 5 candidate genes can be considered a hub, in fact, we can classify them as focused leaves. For instance, LRRC52 interactions are focused on the PPI layer where his gene is indeed not a hub. Among the genes that we identified (see Table 4.1), the one with a more affluent multiplex role is HA02 with a P_i of 0.97, therefore playing a paper in the most layers, and the gene with the higher o_i, is UBE2X. Both of these genes are classified as multiplex leaves, in addition to CYP3A4 and APOF.

The observation that the interactors of the genes with the highest AS-DE correlations are layer-specific and peripheral opens up interesting hypotheses on the role of local structural properties of multilayer networks on the propagation of the expression dysregulation. Based on these results, we expect less central and less participatory nodes to be more susceptible to failures of the tight expression regulation that characterizes biological systems, in agreement with the increased resilience observed in networks with high density and high heterogeneity [76].

As the genes with the highest AS-DE correlations seem to have a similar structural role in the multilayer network, we plotted both metrics against the AS-DE ρ to assess if these differences have an impact on the results. As presented in Figure 4.8b, there is no apparent relationship between these variables.

Assessing the optimal multilayer combination

To validate the optimality of our selected multilayer network, we performed a run-through of the workflow using all possible combinations of the layers from this study (metabolic reactions, PPI in the liver, and pathways). As presented in Figure 4.9, we obtained discrete results between the different combinations, with poorer results when using single-layer and two-layer networks, compared to using the three layers, where, as previously mentioned, we can identify 41 significantly correlated genes, including 3 strongly correlated. On the other hand, when using one layer, we identify 1, 10, and 5 genes, and with two layers, we identify 14, 13, and 38 genes. With these alternative approaches, we identified three genes with a strong ρ: ADH1C when using metabolic reactions, CHRNA1 with pathways, and TBX5 when combining metabolic reactions and pathways, which has recently been found in tumors with mesenchymal samples [77].

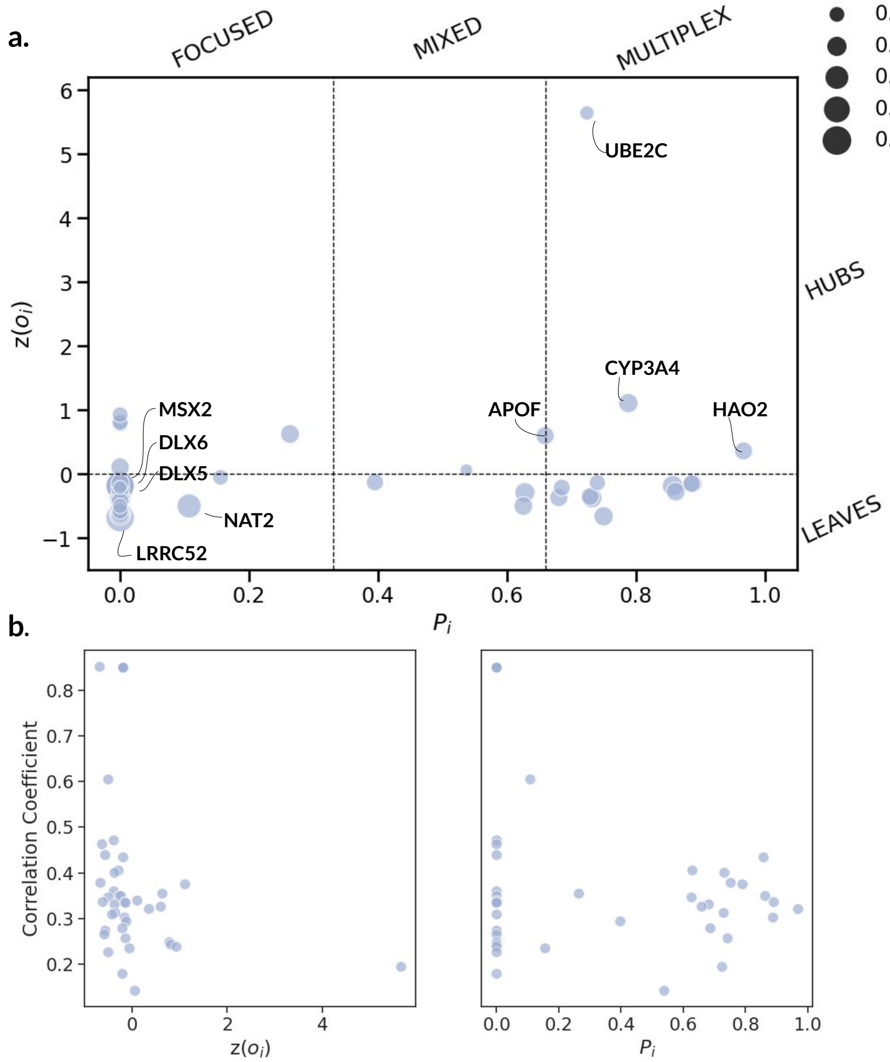

Figure 4.8: Study of the roles of the nodes in the multilayer network (a) Cartography of the multiplex participation coefficient P_i, versus the Z-score of the total overlapping degree $z(o_i)$, highlighting the top 5 candidate genes: LRRC52, DLX5, DLX6 MSX2, and NAT2, and the genes classified as multiplex hubs: UBE2C, CYP3A4, APOF, and HA02. (b) Correlation analysis of the AS-DE ρ against the P_i, $z(o_i)$, and community size.

Pathway enrichment analysis

We completed the exploration of the results by annotating the genes in the prominent communities (see Table 4.1) to cellular pathways, through the KEGG, Reactome, and Wikipathways biological ontologies. When querying the results for the genes in the community of LRRC52, we did not find any enriched pathways, yet, when doing the same with the genes from the second most prominent community, with DLX6, DLX5, and MSX2, we obtained a total of 134 pathways, as shown in Figure 4.10.

From the pathways with a higher gene ratio, count of cases divided by the query size, we can extract that this community is involved in the regulation of RUNX2, runt-

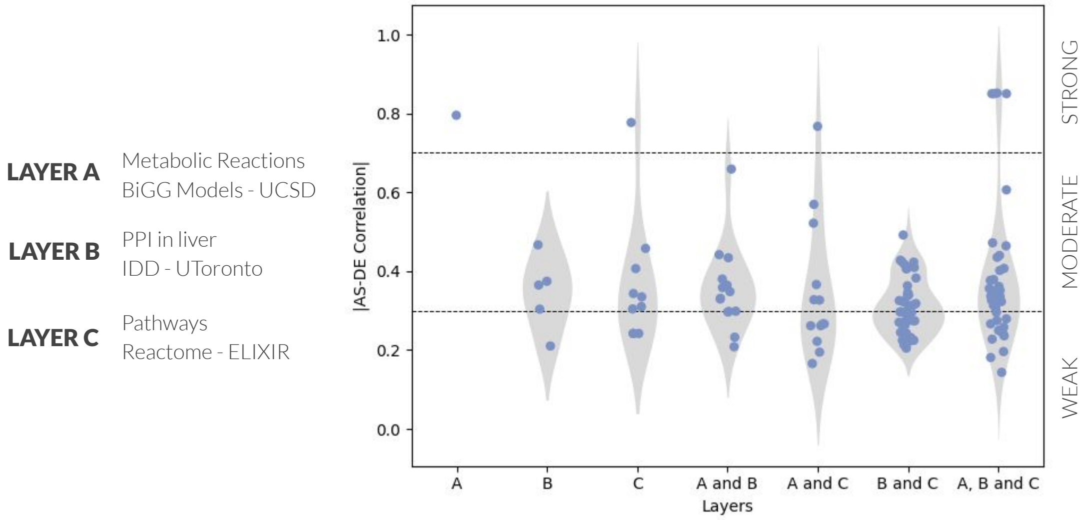

Figure 4.9: Scatterplots and violin plots of the AS-DE correlation results for all single and multilayers network combinations: metabolic reactions, PPI in the liver, and pathways classifying the correlation values for each layer combination into 3 levels: weak (below 0.29), moderate (between 0.3 and 0.69), and strong (between 0.7 and 1)

related transcription factor 2, with DLX5, DLX6, MSX2, and also PSMD4 and PSMD8 genes involved. Which is a transcription factor that plays an important role in different biological processes of osteoblast and chondrocyte, and has recently been implicated in migratory behavior in different human malignancies, including adult liver cancer [78].

Another interesting output of this analysis is the description of four pathways related to Wnt signaling, including the degradation of AXIN, confirming the fact that HB has a high mutation rate of CTNNB1 gene mutations, which constitutively activates the Wnt/β-catenin signaling pathway. In this case, the genes involved are PSMD4 and PSMD8. Another interesting finding is the description of six pathways related to hedgehog signaling pathway, Gli and Hh, with a dysregulation extensively reported in hepatocellular carcinoma. In HB, it has been reported a hypermethylation of its main inhibitor (HHIP) in 26% of HBs and this leads to the abnormal activation of the Hh pathway. In vitro experiments with a DNA demethylating drug restored HHIP protein expression, increased tumor cell apoptosis, and reduced tumor activity, indicating that abnormal activation of the Hh pathway caused by hypermethylation of the HHIP gene plays an important role in the occurrence and development of HB [79].

4.4 Implementation of the multiAffinity tool

We developed multiAffinity, a command-line workflow that defines how gene dysregulation propagates on a multiplex network provided by the users, as shown

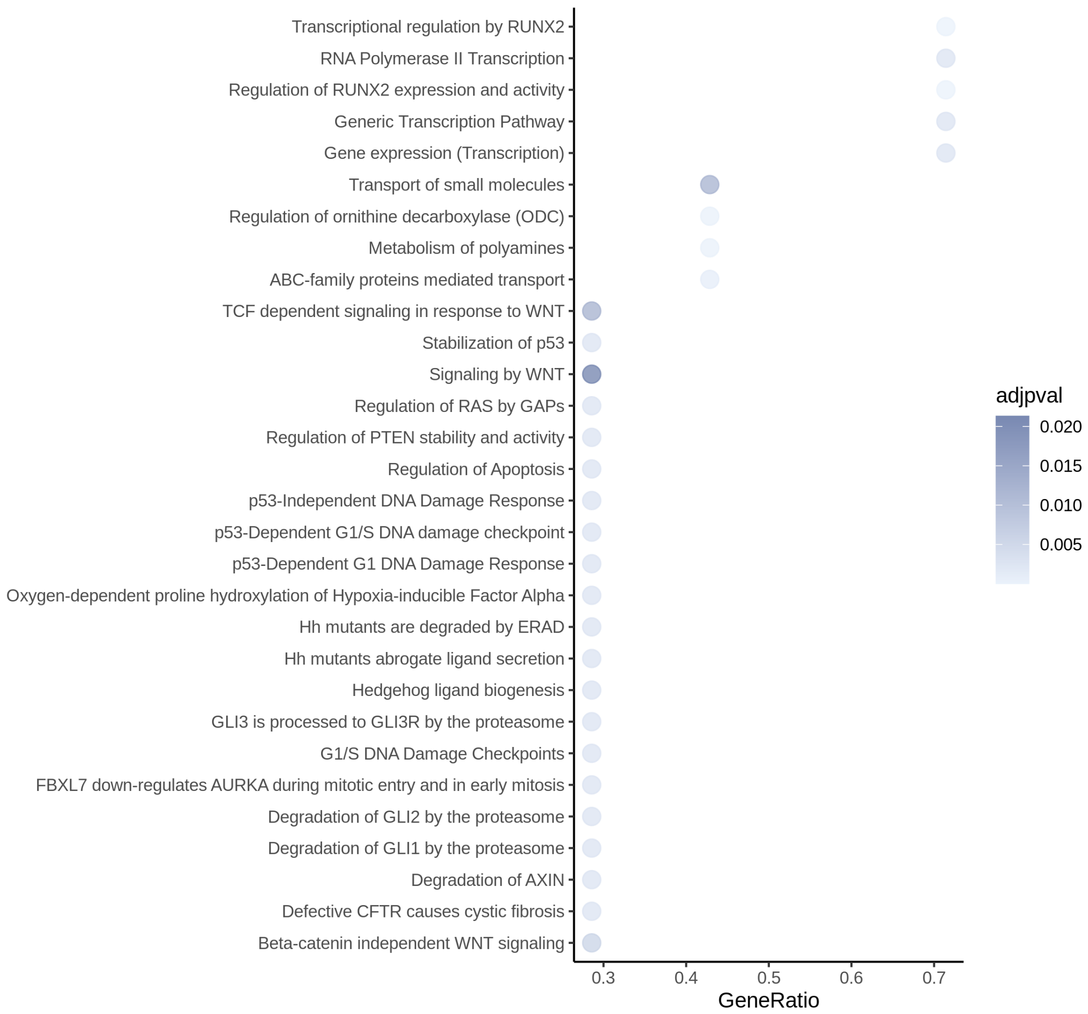

Figure 4.10: Dotplot showing the 30 most significantly enriched pathways. It depicts the adjusted enrichment scores (adjpval) and gene ratios.

in Figure 4.11a. The corresponding source code is freely available at Github (https://github.com/marbatlle/multiAffinity).

The multiAffinity script accepts as input a series of comma-separated values (.csv) files containing the expression data (raw count matrices and metadata) and the network layers, describing interactions between genes. At the end of the execution, it creates an output folder where the gene dysregulation diffusion results are stored as a .csv file. In addition, it generates three folders with the secondary outputs, including information about the DEGs obtained for each study, the significance of the metaDEGs, the heterogeneity across the studies, the defined community structure, and the affinity scores matrix and multiplexity of the multilayer network.

Before running the script, the user can specify a series of parameters, shown in Table 4.2, to customize the job, including whether the pipeline computes the AS-DE correlation on each community (local approach) or with respect to all genes (global approach).

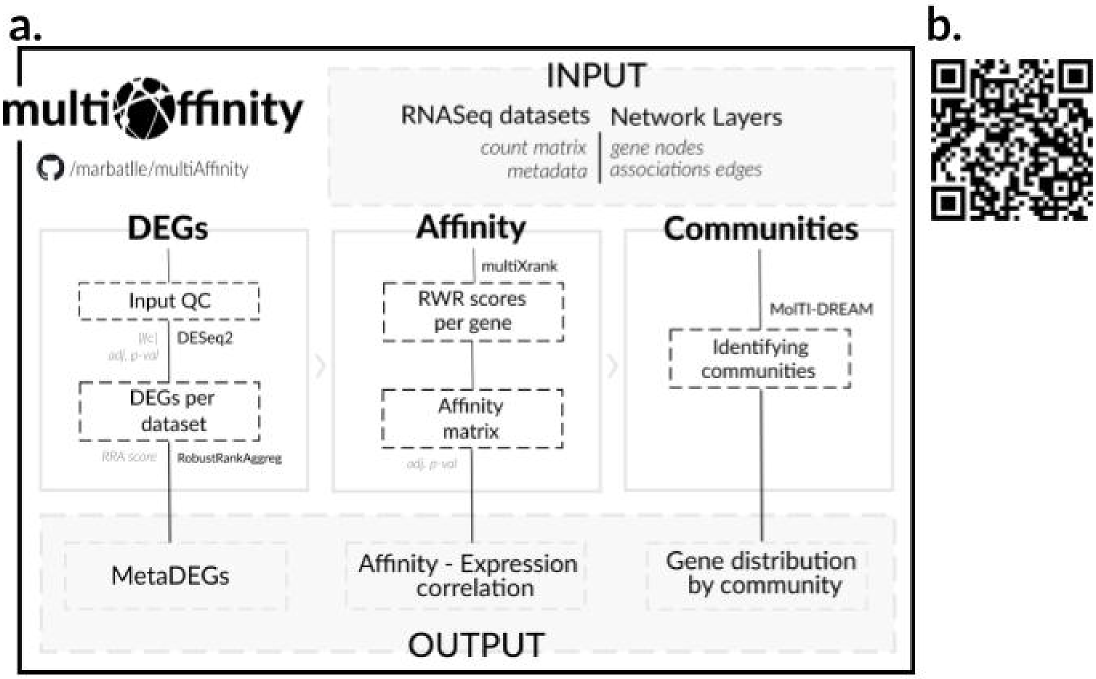

Figure 4.11: Details for the multiAffinity tool: (a) Workflow and (b) iPC-VRE sample demonstration (https://youtu.be/-KFXSQ5NUmo).

Table 4.2: User parameters for the multiAffinity tool settings.

Arguments	Default
Approach	local
Adjusted p-value	0.05
DESeq2 - LFC cutoff	1
multiXrank - R value	0.15
multiXrank - Selfloops	1
MolTI-DREAM - Modularity	1
MolTI-DREAM - Louvain	5

Moreover, the user can set the metadata labels, and define a series of hyperparameters used by the tools incorporated in the pipeline such as the adjusted p-value cutoff and the LFC for the DESeq2 package, the global restart probability (r) and whether self-loops are removed by multiXrank, as well as the Newman modularity resolution parameter γ and the number of randomizations for the MolTI-DREAM software. For more details on the script usage, additional information can be found in the package documentation page (https://marbatlle.github.io/multiAffinity).

To ensure the highest degree of reproducibility with Linux, Windows and MacOS operating systems, we containerized the workflow using the Docker (https://hub.docker.com/repository/docker/marbatlle/multiaffinity) repository. The package can also be implemented through cwltool and run directly from de iPC-VRE while using the iPC private data catalog or user uploaded data (https://vre.ipc-project.bsc.es/openvre/launch/), as demonstrated in Figure 4.11b.

Chapter 5

Discussion

Childhood cancer is a rare disease [29] that unfortunately continues to have a great impact, being the leading disease-cause of death in children in developed countries [30]. Diagnosis spreads across different types, hepatoblastoma (HB) being one of the rarest and least characterized [33], with a largely unknown origin due to its low rate of somatic mutations compared to adult liver cancer. Research efforts to improve this knowledge face unique challenges caused by its low prevalence, resulting in a small size of the patient population, and high research costs. The most common alterations found in HB are activating mutations of the β-catenin encoding gene, CTNNB1 [39], with an incidence of 70%, and NFE2L2, present in approximately 10% of the cases [40].

Our first objective, therefore, was to characterize the current landscape of high-throughput transcriptomics data in HB and to increase the statistical strength of the relevant and available public data. In particular, we focused on available RNA-Seq experiments, which, despite the lack of large sample groups, allow us to perform a comprehensive integration and meta-analysis. The relevance of this approach resides in the fact that genes dysregulated constitutively across different datasets are most likely to pinpoint key biological processes affected in HB.

In the present study, we included five transcriptomic datasets, selected by the IGTP Children's Liver Oncology group: GSE133039, GSE89775, GSE104766, GSE151347, and GSE81928, including 32, 10, 25, 11, and 23 case samples. We integrated these five HB datasets using a standard and robust method, robust rank aggregation (RRA), and computed a 2-Wasserstein distance test to determine that the collected datasets had homogeneous distributions, without a significant inter-study batch effect. This analysis revealed 323 differentially expressed genes (DEGs), with 199 up-regulated and 124 down-regulated genes. RRA, besides being parameter-free and robust to outliers, noise, and errors, is easy to compute and does not require each gene to be present on all datasets. Our approach only used RNA-Seq datasets, but it can be applied to data from different platforms, including microarray datasets, for which the robust rank aggregation method was originally developed [45].

In the context of rare diseases, bioinformatic developments show the capability of molecular networks to provide insights into complex systems and how they can reveal informative patterns within these systems, such as the spread of disease-associated perturbations [13]. In this work, we wanted to take advantage of the increasing availability of differential gene expression data in HB by studying the results of a meta-analysis on a multilayer network representing various levels of molecular information, aiming to uncover the key genes involved in the spreading of dysregulation, and therefore, identify novel candidate genes for further studies.

In previous literature, it was found that disease genes are not scattered randomly in molecular networks, but instead agglomerate in disease modules [16], which can help organize and prioritize disease-associated genes. Therefore, we centered our study of dysregulation spreading using a local approach focused on well-defined multilayer communities, as framed in our second objective of the project. Our multilayer network consisted of 3 layers containing over 1.5 million interactions, defining metabolic reactions, PPIs in liver and pathways, with densities of 0.033, 0.0084, and 0.016, and with the most connected genes being SLC3A2, TMEM27, SLC6A18, YWHAE, APP, TRIM25, RPS27A, UBA52, and UBB. We used the MolTI-DREAM community detection software and filtered the communities to contain a minimum of 7 nodes, to define the disease modules, and after iterating through different modularity resolution parameters (γ), we established an optimal range, $\gamma \in$ [0-15], spanning from 90 communities with an average size of 156.80 to 1420 communities with a size of 10.82 genes.

After establishing the optimal range of γ, to rank the metaDEGs by their involvement in gene dysregulation diffusion, we determined the closeness of the genes using the random walk with restart algorithm (RWR), which yielded affinity scores (AS) for each of the DEGs obtained in the transcriptomics meta-analysis, towards the rest of the genes in the corresponding multilayer communities. By defining subgraphs for each disease module, we constructed a matrix with the AS scores for each metaDEG in the community, and computed the Spearman's correlation between each of these vectors and the correspondent differential expression values (LFC), obtaining a ρ for each metaDEG, from 0 to 1, with higher values indicating a strong AS-LFC correlation.

By running through this multi-step correlation analysis, while iterating over the different community structure configurations set up using the optimal γ range, 0-15, we fulfilled the third aim of this project. Determining that the best results were obtained with 418 disease modules, with an average of 36.76 genes, when setting the γ to 8. With these parameters, we obtained a total of 41 candidate genes with a p-value < 0.05, with 4 genes showing a strong correlation: LRRC52, DLX6, DLX5, and MSX2.

LRRC52 has been related to the regulation of the calcium-activated potassium channels [62], expression in embryonic tissues and stem cells have been located in epithelial cells and the liver [64]. DLX6 and DLX5 encode for members of homeobox transcription factor gene family and work together for the proper craniofacial, axial, and appendicular skeleton development [66]. MSX2 encodes a member of the muscle segment homeobox gene family, which is a transcriptional repressor, with previous observations implying that its normal activity may establish a balance between survival and apoptosis of neural crest-derived cells required for proper craniofacial morphogenesis [72]. None of the selected genes have been directly related to HB, but LRRC52 has been described as an important gene connected with Lymph Node Carcinoma [64], and an important paralog, LRIG1, is a prognostic marker in hepatocellular carcinoma [66]. Further, DLX5 and MSX2 have been shown to be connected with the Wnt/β-catenin pathway, which has been described as the most common alteration in HB [71].

To gain an understanding of the results, we analyzed the multiplicity of the candidate genes, we used the total overlapping degree and the multiplex participation coefficient to differentiate among different categorizations of nodes, namely focused, mixed, and multiplex, and hubs and leaves. The 5 genes with the highest AS-DE correlations, LRRC52 DLX5, DLX6, and MSX2, are focused leaves, meaning that most of their connections are distributed in only one layer and are scarcely linked to other nodes. LRRC52 is focused on the PPI in the liver layers, and DLX5, DLX6, and MSX2 are focused on the pathways layer. We also performed a robustness analysis on the use of our multilayer network, composed of three layers, over the use of single and two-layer networks, obtaining poorer results when using these alternative combinations. As we added more layers, we identified more genes with a significant AS-DE correlation, hence confirming that by integrating different layers, we can understand better interactions that may otherwise not be uncovered. To complete the exploration of the results, we queried the pathways of the most prominent community, which includes DLX5, DLX6, and MSX2, using three biological databases: KEGG, Reactome, and Wikipathways. We uncovered that this community is highly involved in the regulation of RUNX2, which promotes hepatocellular carcinoma cell migration, and we also confirmed that this community of genes is linked to the activation of the Wnt signaling pathway, related to HB in recent literature [78]. Further characterization of these highlighted genes should be performed with experimental examinations, such as through a gene knockout technique, to confirm that these genes are in fact linked to HB and responsible for the inferred dysregulation effects upon the members of their multilayer communities.

We achieved the last goal of the project by deploying the workflow as a code-free tool to the iPC-VRE to enable multi-disciplinary scientists to use this tool to reveal novel candidate cancer genes, with a special focus on its application to rare diseases.

Chapter 6

Conclusions

In this work, we designed a workflow to accurately and robustly analyze the dysregulation signal diffusion in a multilayer network and employed it to determine novel candidate cancer genes in HB, a rare pediatric tumor, where molecular data is scarce. By using this approach, we uncovered 4 potential cancer genes for HB, LRRC52, DLX6, DLX5, and MSX2. Further analysis of the results should focus on experimentally determining the implication of the revealed candidate cancer genes, for example, by performing a gene knockout technique to investigate their functioning and influence in gene expression dysregulation. In addition, we deployed this multistep bioinformatics analysis as a code-free tool to the iPC-VRE, enabling other scientists to apply this analysis to other pediatric cancers. Future steps of this study could include the adaptation of the workflow to develop novel tools for the analysis of other network topological metrics.

Chapter 7

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